Felodipine ER PopPK: model testing log

What was tried, what happened, and why M5 is the final model (M3 fallback)

2026-10-25

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Summary

The PopPK model has to reproduce the PBPK (PK-Sim) bioequivalence decision across the 27 release formulations (Weibull dissolution T50 × shape). Twelve structural variants were tested in three rounds:

1. **Disposition and absorption structure** (M1 ⟶ M4, 2026-10-16 to 10-23): a gut absorption step between release and plasma (M3) cut the T/R Cmax GMR error from up to 20 % (M1) to ≤ 3.5 %.
2. **Delay-style `@delay` input** (M1d–M3d and no-lag variants, 10-24): with the release lag kept, `@delay` reproduces the ODE models exactly; without the lag, fits and curves are much worse.
3. **Fixing M3′s early peak** (M3L, M3Lo, M5, M6, 10-24 to 10-25): a dual-Weibull release (M5) fits better on every arm and cuts the Tmax bias in half; estimating the lag or adding a transit chain does not.

Final model: M5 (dual Weibull in the `@delay` form, fixed lag ⟶ ka ⟶ 2-compartment, fitted per arm). **Fallback: M3.** Equations and Pumas code: `model_equations/felodipine_poppk_models.pdf`.

Common set-up

Item	Setting
Data	27 PK-Sim arms × 100 subjects, 18 samples 0.5–96 h, 15 % proportional noise (seed 20261016); the same noisy dataset for every model
Dose, units	10 mg = 26 024 nmol; h, L, nmol/L
Release lag	0.3113 h, fixed (PK-Sim value) unless stated
IIV	Log-normal on release time, CL, Vc
Residual error	Proportional
Estimation	FOCE, Pumas, per arm; two starting points, the higher LL kept
Diagnostic arms	4.78 h/0.764 (reference), 4.78/0.5, 4.78/1.5, 2.39/0.764, 9.56/0.764
Benchmarks	Cmax and AUC vs PBPK in the same subjects; T/R Cmax GMR vs the reference arm; Tmax and mean curve vs noise-free PK-Sim; BE limits and Type I error (felo_02–04)

All models at a glance

Diagnostic arms, best start per arm. LL is the log-likelihood (higher is better); GMR error is the model's T/R Cmax GMR against PBPK's; Tmax bias is the median per-subject Tmax minus PK-Sim's.

Model	Structure	Par.	Best LL, 5 arms (ref, 0.5, 1.5, 2.39, 9.56)	GMR error	Tmax bias	Mean-curve error	Outcome
M1	Weibull → 1-cmt	8	−557, −786, −366, −719, −395	−20 to −9 %	−1.5 to +0.35 h	3.8–12.9 %	Baseline; rejected
M2	Weibull → 2-cmt	10	−557, −593, −247, −603, −388	−11 to −6 %	−1.2 to +0.3 h	1.9–10.9 %	Peripheral cmt not identified; rejected
M3	Weibull → ka → 2-cmt	11	−354, −487, −59, −481, −108	−3.5 to +3.1 %	−0.7 to +0.1 h	2.0–6.6 %	**Fallback**
M4	M3, reference + test paired	13	not comparable (two arms)	−12 to +15 %	—	—	Rejected
M1d / M2d	M1 / M2 in `@delay` form	8 / 10	as M1 / M2 (within 0.3)	as M1 / M2	as M1 / M2	as M1 / M2	Equivalence check passed
M3d	M3 in `@delay` form	11	−354, −488, −56, −481, −107	as M3	as M3	as M3	Equivalence check passed
M1d0 / M2d0 / M3d0	`@delay`, no lag	8 / 10 / 11	M3d0: −755, −754, −48, −866, −500	M3d0: −10.4 to 0 %	M3d0: −0.4 to +0.55 h	M3d0: 2.9–8.4 %	Rejected: lag needed
M3L	M3d, lag estimated	12	failed (Inf) on 4 arms; −224 at 1.5	—	—	—	Not estimable
M3Lo	M3 ODE, lag estimated	12	−526, −762, −329, −808, −106	—	—	—	Lag ≈ 0.3 h; worse fits
M5	**Dual Weibull → ka → 2-cmt**	**14**	**−348, −439, +38, −433, −85**	**−3.3 to 0 %**	**−0.3 to +0.15 h**	**1.2–4.6 %**	**Final**
M6	Weibull → 3 transit → 2-cmt	11	−365, −479, +19, −484, −109	−3.1 to +3.2 %	−0.45 to 0 h	2.1–4.3 %	Partial fix; not selected

Round 1: disposition and absorption structure (2026-10-16 to 10-23)

M1, 1-compartment + Weibull release. The first production model, fitted on all 27 arms. It had two likelihood basins, and in 23 of 27 arms the best fit was a fast "release" (implied T50 0.3–2.4 h) with a very large Vc (≈ 4000 L) — release and distribution were not separated. Cmax/PBPK ranged 0.87–1.135 across arms, which biased the T/R Cmax GMR by up to 20 %. The PopPK safe space diverged from PBPK away from the reference shape.

M2, 2-compartment + Weibull release. Q or Vp ran to their bounds from either start: the peripheral compartment is not identified without an absorption step. Rejected.

M3, Weibull release → first-order absorption (ka) → 2-compartment. The gut compartment separates release from appearance in plasma, giving PBPK's fast rise and plateau without forcing release to be fast. On all 27 arms: AIC better than M1 on 27/27, Cmax/PBPK 0.976–1.043, GMR error ≤ 3.5 % (M1: 20 %). Remaining problems: the fitted T50 is about twice PK-Sim's (≈ 9.4 h at the reference 4.78 h), Vc is non-physiological on some arms (≈ 2500 L at shape 1.5), and the curve peaks 0.4–0.7 h early at shape ≥ 0.764.

M4, M3 paired (reference and test in the same subjects, shared disposition). Fitted to two arms at once, with separate release parameters per period. Much worse than two separate M3 fits; several fits produced non-finite predictions for up to 75 of 100 subjects, and the GMR error was −12 % to +15 %. Reason: in the PBPK model the release rate changes where along the gut the drug is absorbed and so how much gut-wall CYP3A4 extraction it sees, so disposition as seen from the gut is not shared across formulations. Rejected.

Round 2: delay-style `@delay` input (2026-10-24)

Question. Do the results stay the same if the hand-written Weibull hazard ODE is replaced by the Pumas built-in delay form (`@delay(Weibull(shape, scale), cmt)` with the dose at bioavailability 0)?

What happened.

- The first version had no lag, as in the plain `@delay` form. Fits were much worse (M3d0 reference LL −755 vs −354), M1/M2 peaked too early, and M3d0 overshot by 60 % at 0.5 h.
- Adding the lag inside `@delay` needed workarounds:
  - Pumas ignores `@dosecontrol` lags when `@delay` is used (identical output with and without).
  - `Tlag + Weibull` gives an infinite likelihood for shape < 1, as does a dose record placed at t = lag.
  - Working form: `@delay(Tlag + truncated(Weibull(γ, α), 1e-4, Inf), cmt)`. The truncation removes at most 0.4 % of the release (shape 0.5), less than the 0.01 h clock floor already used in M3.
- **With the lag, M1d/M2d/M3d reproduce M1/M2/M3**: LLs within 3 units, identical Tmax and mean curves, same GMRs. Run times were similar to the ODE.

Conclusion. The `@delay` form is equivalent and was adopted for the new structures; the lag must stay.

Round 3: fixing M3's early peak (2026-10-24 to 10-25)

Diagnosis. M3 rises correctly but peaks early and falls too fast, worst at shape 1.5 (Tmax 5.3 vs 6.0 h, only 28 % of subjects within 0.5 h of PK-Sim). Two physiological explanations were checked first:

- *Regional gut-wall extraction varying with release rate*: rejected — AUC per fraction absorbed is flat within ±3 % across the 27 arms.
- *Deconvolution against PK-Sim's "absorbed into mucosa" output*: inconclusive (implied half-lives ≤ 2 h).

The remaining explanation is structural: one release timescale and one ka cannot give both the fast initial phase and the long late tail. Four variants were tested on the diagnostic arms:

Variant	Idea	What happened
M3L	Estimate the lag (`@delay`)	Infinite likelihood at the start on all 4 arms with shape < 1 (the lag derivative of the truncated density is singular); at shape 1.5 LL −224 vs −56 fixed
M3Lo	Estimate the lag (ODE)	Lag lands at 0.27–0.33 h on most fits (≈ the fixed 0.3113); LL worse than fixed on 4/5 arms; the second start ended far worse (LL −840 to −5650) on every arm
M5	**Two Weibull release components (FF mixture)**	**Best LL on all 5 arms (+7 to +97); Tmax bias −0.3 to +0.15 h; mean-curve error 1.2–4.6 %**
M6	3 transit compartments instead of ka	Fixes the shape-1.5 curve (LL +19) but no better than M3 elsewhere; 2 of 10 fits failed

Decision (2026-10-24): M5 final, M3 fallback.

M5 on all 27 arms (2026-10-24/25)

Fitted with `02c_fit_poppk_m5.jl` (results in `output_rich_m5/`), compared with M3 by `felo_09`.

Measure	M3 (fallback)	M5 (final)
AIC better	0 / 27 arms	**27 / 27** (median ΔAIC −48, range −289 to −6.5)
Cmax ratio PopPK/PBPK	0.976–1.043	**0.976–1.019**
$AUC_{0\text{-}t}$ ratio PopPK/PBPK	0.985–1.012	**0.989–1.004**
Cmax T/R GMR error vs PBPK	max 3.5 %, mean 1.6 %	**max 3.3 %, mean 1.3 %**
Tmax bias (median of arms)	0.45 h	**0.20 h**

Measure	M3 (fallback)	M5 (final)
Subjects with Tmax within 0.5 h	64 %	**76 %**
2-D safe-space T50 limits vs PBPK	mean |Δ| 0.08 h, bias +0.03 h	mean |Δ| 0.09 h, bias −0.09 h

Caveats.

- The two release components describe the curve; they are not two physical processes (fast fraction 0.02–0.97 across arms; on some arms component a is a 0.2–0.4 h burst).
- The mixture's 50 %-release time is ≈ 8–9 h at PK-Sim 4–5 h, the same offset as M3: the PopPK input lumps release, gut transit and absorption.
- At the reference arm the fast fraction (RSE 45 %) and Q, Vp (RSE 55–70 %) are poorly determined; Vc reaches ≈ 2300 L at shape 1.5, as in M3.
- One arm (6.31 h / 1.2) still peaks 0.85 h early.
- The 2-D safe-space limits sit slightly low because M5 overpredicts the reference-arm Cmax by 0.9 %, so its T/R GMRs run about 1.5 % low at shape ≥ 0.9.

BE decision vs PBPK (felo_04, 2026-10-25)

T50 axis (shape 0.764): limits where BE is lost (h). Cmax sets every limit; AUC never reaches it.

	Lower	Upper
PBPK	3.25	6.96
M3	3.40 (+0.14)	6.83 (−0.13)
M5	**3.27 (+0.02)**	**6.86 (−0.10)**

Type I error — chance of passing BE when the true (PBPK) product sits at the limit the PopPK model would declare (nominal 5 %; 20 000 simulated trials, seed 1234):

	M3	M5
Lower, Cmax, n = 24 / 54	7.6 % / 10.2 %	**5.3 % / 5.8 %**
Upper, Cmax, n = 24 / 54	5.8 % / 6.8 %	**5.6 % / 6.3 %**
Lower, joint, n = 24 / 54	5.8 % / 10.1 %	**4.0 % / 5.6 %**
Upper, joint, n = 24 / 54	4.6 % / 6.6 %	**4.4 % / 6.0 %**

Shape axis (T50 4.78 h). PBPK loses BE at shape 1.465. Both PopPK models underpredict the shape-1.5 Cmax GMR (M3 1.235, M5 1.227, PBPK 1.269), so neither reaches 1.25 within the tested range: both would call shapes 1.465–1.5 bioequivalent. Unchanged from M3: the TOST control (reference × 0.80) gives 4.7–5.1 %, and RSABE / ABEL accept 20–22 % / 13–14 % at the ABE limit (n = 24).

Noise check (10-25). To see whether the shape-1.5 shortfall is noise, M5 was refitted on the reference, shape 1.2 and shape 1.5 arms (T50 4.78 h) with two new residual-noise datasets (seeds 101 and 202):

Cmax GMR vs reference (n = 100)	Shape 1.2	Shape 1.5
PBPK	1.119	1.269
M5, seed 20261016 (original)	1.101	1.227
M5, seed 101	1.136	1.242
M5, seed 202	1.112	1.240

At shape 1.2 the seeds land on both sides of PBPK, so that gap is noise (about ±1.5 %). At shape 1.5 M5 is 2–3 % low and below 1.25 on every seed: the shortfall is structural. The fitted fast fraction at the reference arm varied from 0.02 to 0.15 between seeds, so the two release components describe the curve rather than being identified individually.

Run notes

Date	Event
10-23	A queued run never started: the `pgrep -f` waiter matched its own command line (~7.5 h lost). Waiters now use the PID or a sentinel file.
10-24	A finished M4 run looked hung: its wrapper was a zombie, which `kill -0` still reports as alive.
10-24	M2d0 at the reference arm (disposition start) hung > 40 min and was killed; recorded as skipped.
10-24	Pumas `@delay` lag limitations found (Round 2).
10-24/25	M5 27-arm run: two scripting bugs fixed in the first hour (arm ordering; loading saved fits needs the original module name). At ~58 min per arm with two starts, the run was switched to the absorption-limited start only — on the 7 arms fitted from both it was never worse. 27/27 arms fitted; about 11 h wall time.
10-25	Noise check (`scripts/run_m5_noise_replicates.sh`, seeds 101/202, 3 arms each): 6/6 fits converged, 10–29 min each, done 16:14 IST.

Files

What	Where (under `Felodipine-BCSII/poppk/`)
Equations and Pumas code, M1–M6	`model_comparison/model_equations/felodipine_poppk_models.pdf`
Short model notes	`model_comparison/MODELS.md`
Diagnostic-arm tables	`model_comparison/diagnostic_*.csv`, `delay_vs_ode_*.csv`
Tmax and mean curves vs PK-Sim	`model_comparison/curve_tmax/`
M5 vs M3, all 27 arms	`model_comparison/m5_vs_m3_all27/`
M5 results, BE sweep, BE comparison	`output_rich_m5/`, `output_sweep_2d_poppk_m5/`, `output_rich_m5/compare/`
M3 results (fallback)	`output_rich_v2/`, `output_sweep_2d_poppk_v2/`, `output_rich_v2/compare/`
Pumas sources and fit scripts	`poppk-fits/src/`, `scripts/` (02b M3, 02c M5, 04–06 diagnostics)